

Package ‘manMetaVAR’

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Coefficients

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Compress

Compress Replication

Description

Compress Replication

Usage

Compress(taskid, repid, output_folder)

Arguments

- taskid Positive integer. Task ID.
- repid Positive integer. Replication ID.
- output_folder Character string. Output folder.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Dynamics	<i>Process Dynamics</i>
----------	-------------------------

Description

Process Dynamics

Usage

Dynamics(dynamics)

Arguments

- dynamics 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: [GenData\(\)](#)

Examples

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

FigBias

*Plot Relative Bias***Description**

Plot relative bias for the fixed and random effects.

Usage

```
FigBias(
  results,
  bias = TRUE,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  rm_zero = FALSE,
  ylim = c(-0.15, 0.15)
)
```

Arguments

results	results data frame.
bias	Logical. If bias = TRUE, plot absolute bias. If bias = FALSE, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigBias(results)

## End(Not run)
```

Description

Plot coverage probabilities for the fixed and random effects.

Usage

```
FigCoverage(  
  results,  
  dynamics = 1,  
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),  
  rm_zero = FALSE,  
  ylim = c(0, 1)  
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigCoverage(results)  
  
## End(Not run)
```

FigPower

*Plot Statistical Power***Description**

Plot statistical power probabilities for the fixed and random effects.

Usage

```
FigPower(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  rm_zero = FALSE,
  ylim = c(0, 1)
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
rm_zero	Logical. Remove cases where the population parameter is equal to zero.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigRMSE\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigPower(results)

## End(Not run)
```

FigRMSE*Plot RMSE*

Description

Plot RMSE for the fixed and random effects.

Usage

```
FigRMSE(  
  results,  
  dynamics = 1,  
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),  
  rm_zero = FALSE,  
  ylim = c(0, 0.25)  
)
```

Arguments

<code>results</code>	results data frame.
<code>dynamics</code>	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
<code>method</code>	Character vector. Methods to include in the plot.
<code>rm_zero</code>	Logical. Remove cases where the population parameter is equal to zero.
<code>ylim</code>	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#), [FigType1Error\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigRMSE(results)  
  
## End(Not run)
```

FigType1Error

Plot Type 1 Error Rates

Description

Plot type 1 error rates for zero random effects.

Usage

```
FigType1Error(
  results,
  dynamics = 1,
  method = c("MetaVAR", "SeqVAR", "BMLVAR"),
  ylim = c(0, 0.25)
)
```

Arguments

results	results data frame.
dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
method	Character vector. Methods to include in the plot.
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigType1Error(results)

## End(Not run)
```

FitDTVAR*Fit the Model using the fitDTVARmxD Package*

Description

The function fits the model using the [fitDTVARmxD::fitDTVARmxD](#) package.

Usage

```
FitDTVAR(data, wd = ".", ncores = NULL, seed = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>wd</code>	Character string. Working directory.
<code>ncores</code>	Positive integer. Number of cores to use.
<code>seed</code>	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

Description

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

Usage

```
FitMetaVAR(data, fit, ncores = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  data = data,
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

FitMLVAR*Fit the Model using the mlVAR Package*

Description

The function fits the model using the [mlVAR::mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:  
set.seed(42)  
data <- GenData(taskid = 1)  
fit <- FitMLVAR(data = data)  
print(fit)  
summary(fit)  
  
## End(Not run)
```

FitMplus*Fit the Model using Mplus*

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>chains</code>	Integer. Number of chains.
<code>iter</code>	Integer. Number of iterations.
<code>fscores</code>	Integer. Number of iterations for factor scores.
<code>plot</code>	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.
<code>default_priors</code>	Logical. If <code>default_priors = TRUE</code> , use default priors.
<code>wd</code>	Character string. Working directory.
<code>mplus_bin</code>	Character string. Path to Mplus binary. If <code>mplus_bin = NULL</code> , the function will try to find the appropriate binary.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

GenData

*Simulate Data***Description**

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

`taskid` Positive integer. Task ID.

See Also

Other Data Generation Functions: [Dynamics\(\)](#)

Examples

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

manmetavar-data-methods

*Methods for Objects of Class manmetavar.data***Description**

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class <code>manmetavar.data</code> .
...	additional arguments.
object	Object of class <code>manmetavar.data</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-dtvar-methods

Methods for Objects of Class `manmetavar.dtvvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvvar`.

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```

    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

```

Arguments

object	Object of class <code>manmetavar.dtvar</code> .
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.

nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
...	additional arguments.
x	Object of class manmetavar.dtvvar.
means	Logical. If means = TRUE, return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-metavar-methods

Methods for Objects of Class manmetavar.metavar

Description

This page documents the available methods for objects of class manmetavar.metavar.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```



```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)
```

Arguments

object	Object of class <code>manmetavar.metavar</code> .
...	additional arguments.
x	Object of class <code>manmetavar.metavar</code> .
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
lb	Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mlvar-methods

Methods for Objects of Class `manmetavar.mlvar`

Description

This page documents the available methods for objects of class `manmetavar.mlvar`.

Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mlvar'
summary(
```

```

    object,
    show = c("fit", "temporal", "contemporaneous", "between"),
    digits = 4,
    ...
)

```

Arguments

x	Object of class <code>manmetavar.mlvar</code> .
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.
object	Object of class <code>manmetavar.mlvar</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mplus-methods

Methods for Objects of Class `manmetavar.mplus`

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,

```

```

    burnin = NULL,
    legend_loc = "topright",
    ...
)

```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
median	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
burnin	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.
...	additional arguments.
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
x	Object of class <code>manmetavar.mplus</code> .
what	Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput	<i>Create Mplus Input File</i>
------------	--------------------------------

Description

The function creates an Mplus input file.

Usage

```

MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,

```

```

    fscores,
    plot,
    default_priors,
    ncores = NULL
)

```

Arguments

dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

Examples

```

cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = TRUE,
    ncores = NULL
  )
)

```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

data(params)

Format

A dataframe with 27 rows and 4 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions.

dynamics Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

results	<i>Simulation Results</i>
---------	---------------------------

Description

Simulation Results

Usage

data(results)

Format

A dataframe with I rows and J columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occasions.

dynamics Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p -value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sq_error Mean squared error.

bias Bias.

rel_bias Relative bias.

se_bias Bias in standard error estimate.

rel_se_bias Relative bias in standard error estimate.

rmse Root mean square error.

coverage Coverage probability.

power Statistical power.

par_idx Parameter index.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim

Simulation Replication

Description

Simulation Replication

Usage

```

Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mlvar,
  mplus,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

 SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR	<i>Simulation Replication - FitMetaVAR</i>
---------------	--

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR

Simulation Replication - FitMLVAR

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus

*Simulation Replication - FitMplus***Description**

Simulation Replication - FitMplus

Usage

```

SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3</code> ; to Mplus input file.
default_priors	Logical. If <code>default_priors = TRUE</code> , use default priors.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

- output_type Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output_folder Character string. Output folder.
- suffix Character string. Output of manCTMed:::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj

Simulation Project Name

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	Summary
-----	---------

Description

Summary

Usage

```
Sum(
  taskid,
  reps,
  output_folder,
  overwrite,
  integrity,
  metavar,
  mlvar,
  mplus,
  ncores
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARLB	<i>Summary: Likelihood-Based (FitMetaVAR)</i>
-----------------	---

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR

Summary (FitMLVAR)

Description

Summary (FitMLVAR)

Usage

```
SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus	<i>Summary (FitMplus)</i>
-------------	---------------------------

Description

Summary (FitMplus)

Usage

SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

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